ME101: Engineering Mechanics (3 1 0 8)

2023-2024 (II Semester)

Division II & IV

Instructor

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L05

Structural Analysis

Engineering Structure

An engineering structure is any connected system of members built to support or transfer forces and to safely withstand the loads applied to it.

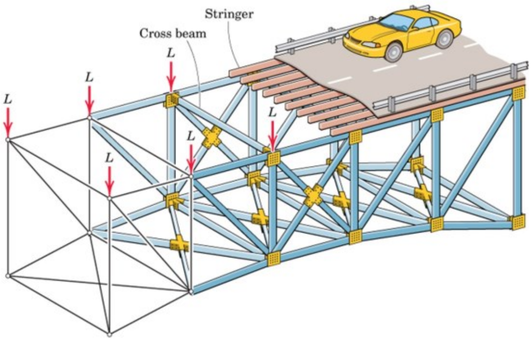
Structural Analysis

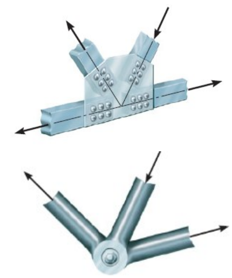
Structural Analysis

ME101→ Trusses/Frames/Machines/Beams/Cables

→ Statically Determinate Structures

To determine the internal forces in the structure, dismember the structure and analyze separate free body diagrams of individual members or combination of members.



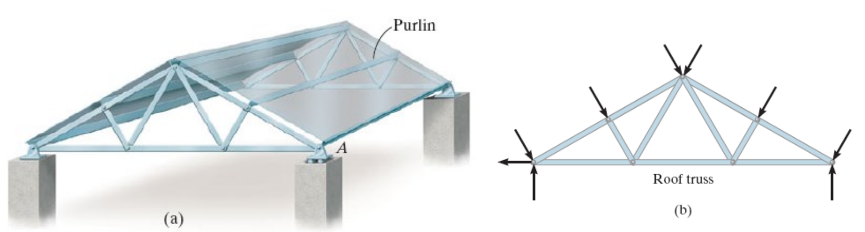


Structural Analysis: Plane Truss

Truss: A framework composed of members joined at their ends to form a rigid structure

– Joints (Connections): Welded, Riveted, Bolted, Pinned

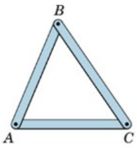
Plane Truss: Members lie in a single plane



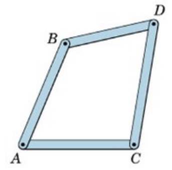
Structural Analysis: Plane Truss

Simple Trusses

Basic Element of a Plane Truss is the Triangle



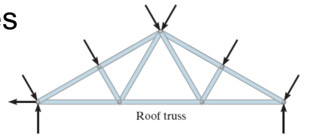
- Three bars joined by pins at their ends → Rigid Frame
  – Non-collapsible and deformation of members due to induced internal strains is negligible

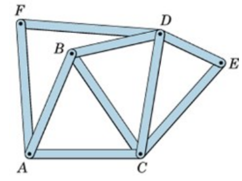


- Four or more bars polygon → Non-Rigid Frame
  How to make it rigid or stable?

  by forming more triangles!

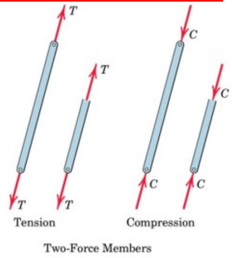
Structures built from basic triangles
→Simple Trusses



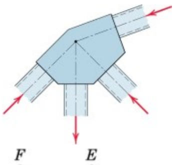


Structural Analysis: Plane Truss

Basic Assumptions in Truss Analysis

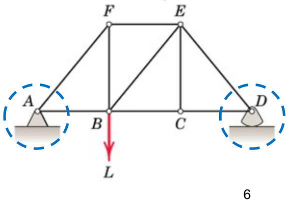


- All members are two-force members
- Weight of the members is small compared with the force it supports (weight may be considered at joints).
  – no effect of bending on members even if weight is considered
- External forces are applied at the pin connections
- Welded or riveted connections → Pin Joint if the member centerlines are concurrent at the joint



Common Practice in Large Trusses

- Roller/Rocker at one end. Why?
  - to accommodate deformations due to temperature changes and applied loads.
  - otherwise it will be a statically indeterminate truss



Structural Analysis: Plane Truss

Truss Analysis: Method of Joints

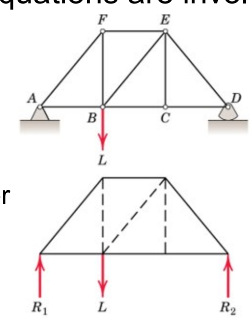
- Finding forces in members

Method of Joints: Conditions of equilibrium are satisfied for the forces at each joint

- Equilibrium of concurrent forces at each joint
- only two independent equilibrium equations are involved

Steps of Analysis

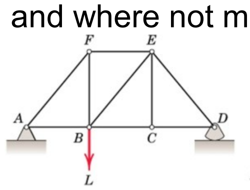
1. Draw Free Body Diagram of Truss
2. Determine external reactions by applying equilibrium equations to the whole truss
3. Perform the force analysis of the remainder of the truss by Method of Joints

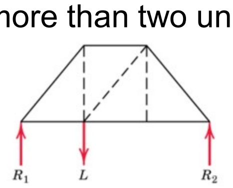


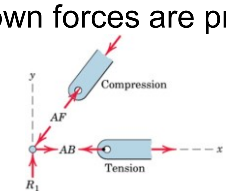
Structural Analysis: Plane Truss

Method of Joints

- Start with any joint where at least one known load exists and where not more than two unknown forces are present.







FBD of Joint A and members AB and AF: Magnitude of forces denoted as *AB* & *AF*

- Tension indicated by an arrow away from the pin
- Compression indicated by an arrow toward the pin

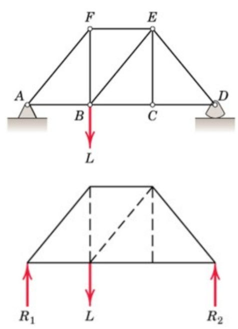
Magnitude of AF from $\Sigma F_y = 0$

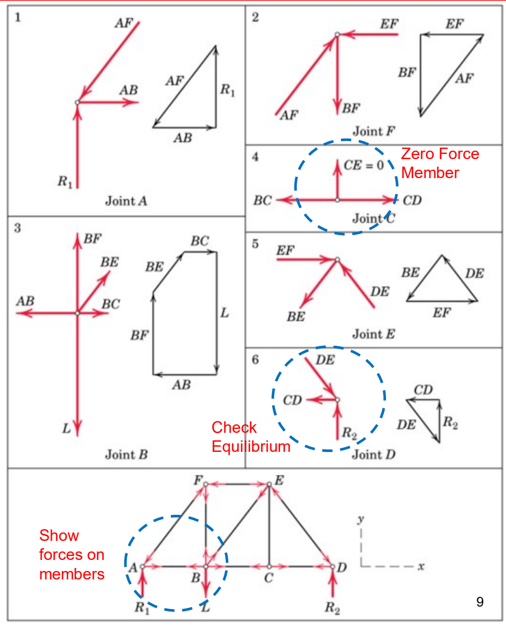
Magnitude of AB from $\Sigma F_x = 0$

Analyze joints F, B, C, E, & D in that order to complete the analysis

Structural Analysis: Plane Truss

Method of Joints

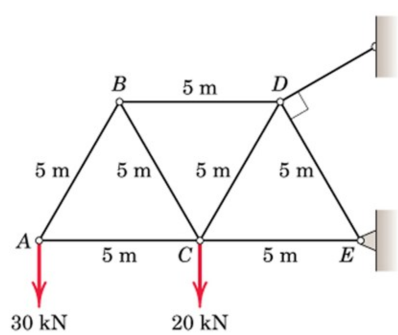




- Negative force if assumed sense is incorrect

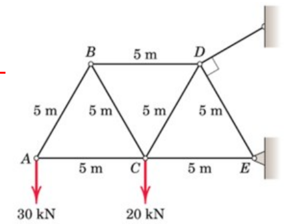
Method of Joints: Example

Determine the force in each member of the loaded truss by Method of Joints

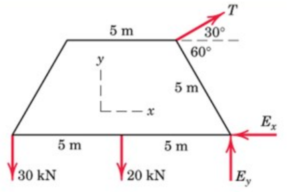


Method of Joints: Example

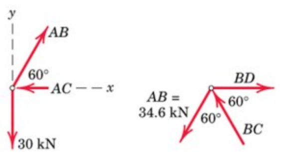
Solution



$[\Sigma M_E = 0]$	$5T - 20(5) - 30(10) = 0$	$T = 80$ kN
$[\Sigma F_x = 0]$	$80 \cos 30° - E_x = 0$	$E_x = 69.3$ kN
$[\Sigma F_y = 0]$	$80 \sin 30° + E_y - 20 - 30 = 0$	$E_y = 10$ kN



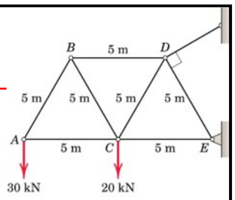
$[\Sigma F_y = 0]$	$0.866AB - 30 = 0$	$AB = 34.6$ kN T
$[\Sigma F_x = 0]$	$AC - 0.5(34.6) = 0$	$AC = 17.32$ kN C
$[\Sigma F_y = 0]$	$0.866BC - 0.866(34.6) = 0$	$BC = 34.6$ kN C
$[\Sigma F_x = 0]$	$BD - 2(0.5)(34.6) = 0$	$BD = 34.6$ kN T

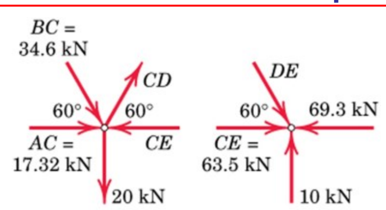


Joint A Joint B

Method of Joints: Example

Solution





Joint C Joint E

$[\Sigma F_y = 0]$ $0.866CD - 0.866(34.6) - 20 = 0$

$CD = 57.7$ kN T

$[\Sigma F_x = 0]$ $CE - 17.32 - 0.5(34.6) - 0.5(57.7) = 0$

$CE = 63.5$ kN C

$[\Sigma F_y = 0]$ $0.866DE = 10$ $DE = 11.55$ kN C

and the equation $\Sigma F_x = 0$ checks.

Structural Analysis: Plane Truss

When more number of members/supports are present than are needed to prevent collapse/stability

→ Statically Indeterminate Truss

- cannot be analysed using equations of equilibrium alone!
- additional members or supports which are not necessary for maintaining the equilibrium configuration → Redundant

Internal and External Redundancy

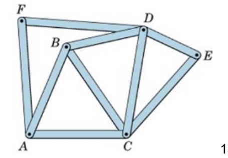
Extra Supports than required → External Redundancy

– Degree of indeterminacy from available equilibrium equations

Extra Members than required → Internal Redundancy

(truss must be removed from the supports to calculate internal redundancy)

– Is this truss statically determinate internally?



Truss is statically determinate internally if $m + 3 = 2j$

$m = 2j – 3$ m is number of members, and j is number of joints in truss

Structural Analysis: Plane Truss

Internal Redundancy or
Degree of Internal Static Indeterminacy

Extra Members than required → Internal Redundancy

Equilibrium of each joint can be specified by two scalar force equations → 2j equations for a truss with “j” number of joints

→ Known Quantities

For a truss with “m” number of two force members, and maximum 3 unknown support reactions → Total Unknowns = m + 3

(“m” member forces and 3 reactions for externally determinate truss)

Therefore:

$m + 3 = 2j$ → Statically Determinate Internally

$m + 3 > 2j$ → Statically Indeterminate Internally

$m + 3 < 2j$ → Unstable Truss

A necessary condition for Stability but not a sufficient condition since one or more members can be arranged in such a way as not to contribute to stable configuration of the entire truss

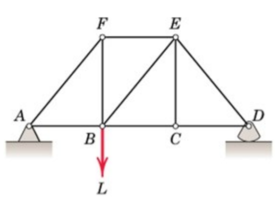
Structural Analysis: Plane Truss

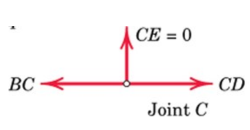
Why to Provide Redundant Members?

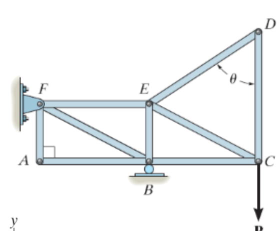
- To maintain alignment of two members during construction
- To increase stability during construction
- To maintain stability during loading (Ex: to prevent buckling of compression members)
- To provide support if the applied loading is changed
- To act as backup members in case some members fail or require strengthening
- Analysis is difficult but possible

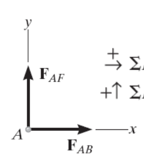
Structural Analysis: Plane Truss

Zero Force Members



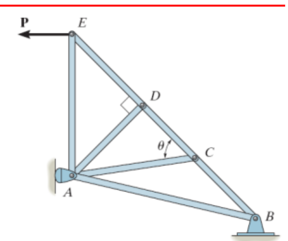


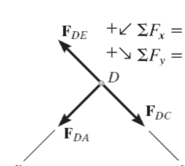




$\xrightarrow{+} \Sigma F_x = 0;\ \boxed{F_{AB} = 0}$

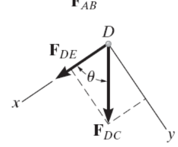
$+\uparrow \Sigma F_y = 0;\ \boxed{F_{AF} = 0}$





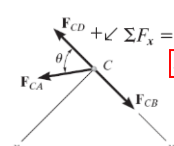
$+\swarrow \Sigma F_x = 0;\ \boxed{F_{DA} = 0}$

$+\searrow \Sigma F_y = 0;\ F_{DC} = F_{DE}$



$+\searrow \Sigma F_y = 0;\ F_{DC} \sin\theta = 0;\ \boxed{F_{DC} = 0}$ since $\sin\theta \neq 0$

$+\swarrow \Sigma F_x = 0;\ F_{DE} + 0 = 0;\ \boxed{F_{DE} = 0}$



$+\swarrow \Sigma F_x = 0;\ F_{CA} \sin\theta = 0$

$\boxed{F_{CA} = 0}$

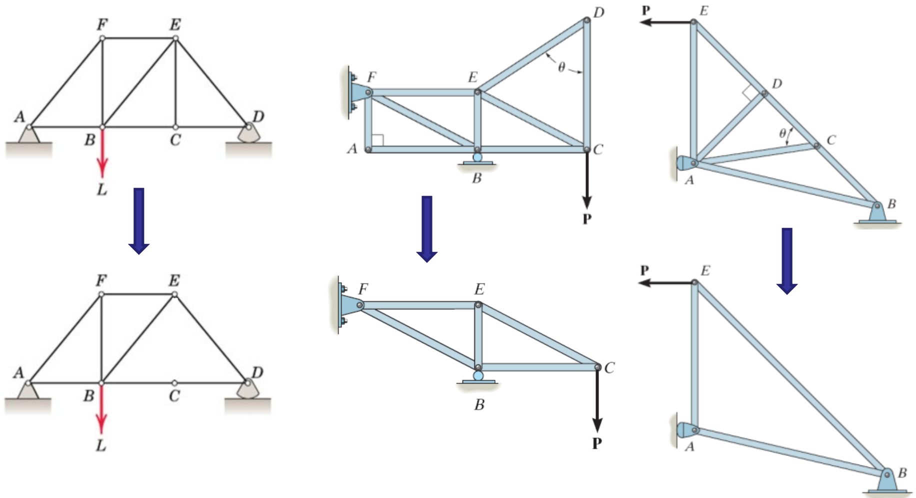
since $\sin\theta \neq 0$

$+\searrow \Sigma F_y = 0;$

$F_{CB} = F_{CD}$

Structural Analysis: Plane Truss

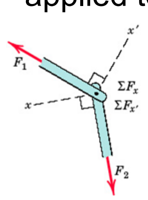
Zero Force Members: Simplified Structures



Structural Analysis: Plane Truss

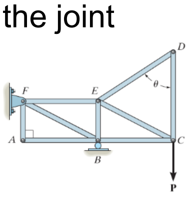
Zero Force Members: Conditions

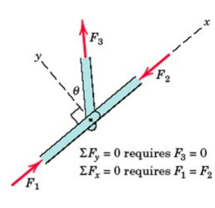
- if only two noncollinear members form a truss joint and no external load or support reaction is applied to the joint, the two members must be zero force members
- if three members form a truss joint for which two of the members are collinear, the third member is a zero-force member provided no external force or support reaction is applied to the joint

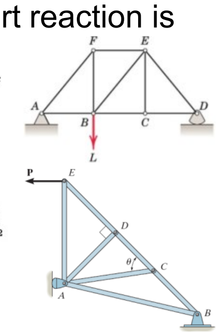


$\Sigma F_x = 0$ requires $F_1 = 0$

$\Sigma F_{x'} = 0$ requires $F_2 = 0$



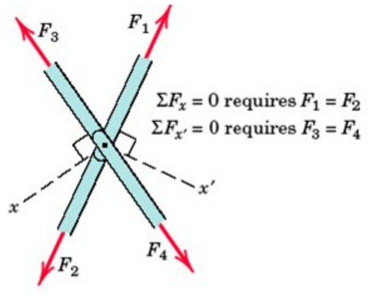




Structural Analysis: Plane Truss

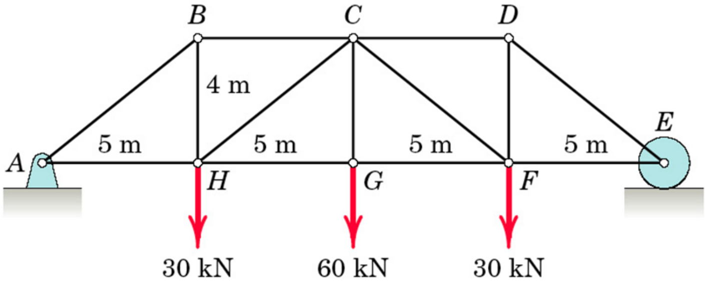
Special Condition

- When two pairs of collinear members are joined as shown in figure, the forces in each pair must be equal and opposite.



Method of Joints: Example

Determine the force in each member of the loaded truss by Method of Joints.

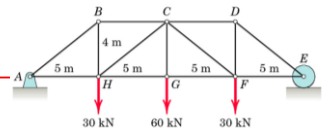


Is the truss statically determinant externally? Yes

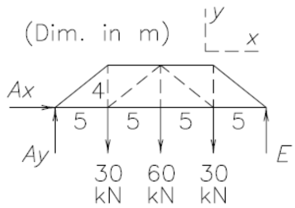
Is the truss statically determinant internally? Yes

Are there any Zero Force Members in the truss? No

Method of Joints: Example



Solution



As a whole: $\Sigma F_x = 0 \Rightarrow A_x = 0$

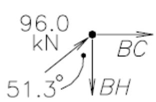
$A_y = E = 60$ kN by

$\Sigma F_y = 0$ and symmetry.

Joint A: $\left(\theta = \tan^{-1}(4/5) = 38.7°\right)$

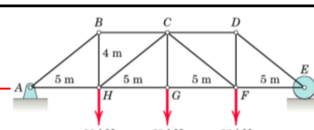
$$\begin{cases} \Sigma F_y = 0 : 60 - AB\sin\theta = 0, \ \underline{AB = 96.0 \text{ kN } C} \\ \Sigma F_x = 0 : AH - 96.0\cos\theta, \ \underline{AH = 75 \text{ kN } T} \end{cases}$$

Joint B:

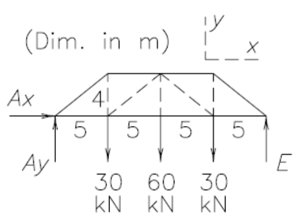


$$\begin{cases} \Sigma F_x = 0 : BC + 96.0\sin 51.3° = 0, \ \underline{BC = -75 \text{ kN } (C)} \\ \Sigma F_y = 0 : -BH + 96.0\cos 51.3° = 0, \ \underline{BH = 60 \text{ kN } T} \end{cases}$$

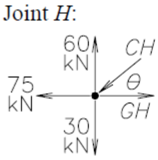
Method of Joints: Example



Solution

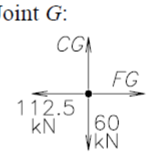


Joint H:



$$\begin{cases} \Sigma F_y = 0 : -CH\sin\theta + 30 = 0, \ \underline{CH = 48.0 \text{ kN } C} \\ \Sigma F_x = 0 : 48.0\cos\theta + GH - 75 = 0, \ \underline{GH = 112.5 \text{ kN } T} \end{cases}$$

Joint G:



$\Sigma F_y = 0 \Rightarrow CG = 60$ kN T

By symmetry:

$FG = 112.5$ kN T, $CF = 48.0$ kN C

$CD = 75$ kN C, $DF = 60$ kN T

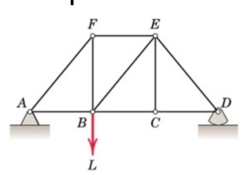
$\underline{EF = 75 \text{ kN } T, \ DE = 96.0 \text{ kN } C}$

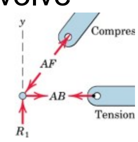
Structural Analysis: Plane Truss

Method of Joints: only two of three equilibrium equations were applied at each joint because the procedures involve concurrent forces at each joint

→Calculations from joint to joint

→More time and effort required



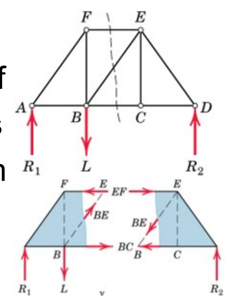


Method of Sections

Take advantage of the 3rd or moment equation of equilibrium by selecting an entire section of truss

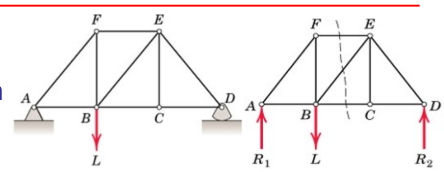
→Equilibrium under non-concurrent force system

→Not more than 3 members whose forces are unknown should be cut in a single section since we have only 3 independent equilibrium equations

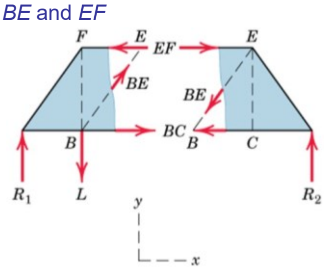


Structural Analysis: Plane Truss

Method of Sections



- Find out the reactions from equilibrium of whole truss
- To find force in member BE:
- Cut an imaginary section (dotted line)
- Each side of the truss section should remain in equilibrium
  - Apply to each cut member the force exerted on it by the member cut away
  - The left hand section is in equilibrium under L, R_1, BC, BE and EF
  - Draw the forces with proper senses (else assume)
    - Moment @ B → *EF*
    - $L > R_1$; → $\sum F_y=0$ → *BE*
    - Moment @ E and observation of whole truss → *BC*
  - Forces acting towards cut section → Compressive
  - Forces acting away from the cut section → Tensile
- Find *EF* from $\sum M_B=0$; Find *BE* from $\sum F_y=0$
- Find *BC* from $\sum M_E=0$



→ Each unknown has been determined independently of the other two

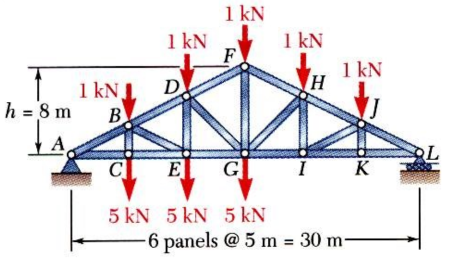
Structural Analysis: Plane Truss

Method of Sections

- Principle: If a body is in equilibrium, then any part of the body is also in equilibrium.
- Forces in few particular member can be directly found out quickly without solving each joint of the truss sequentially
- Method of Sections and Method of Joints can be conveniently combined
- A section need not be straight.
- More than one section can be used to solve a given problem

Structural Analysis: Plane Truss

Method of Sections: Example

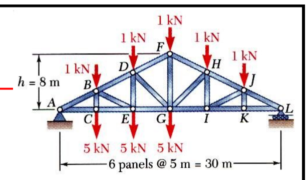


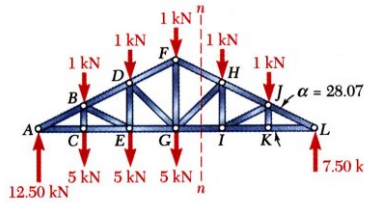
Find out the internal forces in members FH, GH, and GI

Find out the reactions

$$\sum M_A = 0 = -(5\text{ m})(6\text{ kN}) - (10\text{ m})(6\text{ kN}) - (15\text{ m})(6\text{ kN})$$
$$-(20\text{ m})(1\text{ kN}) - (25\text{ m})(1\text{ kN}) + (30\text{ m})L$$
$$L = 7.5\text{ kN} \uparrow$$
$$\sum F_y = 0 = -20\text{ kN} + L + A$$
$$A = 12.5\text{ kN} \uparrow$$

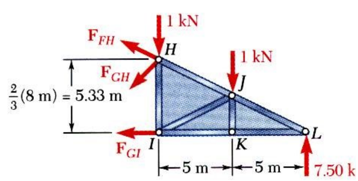
Method of Sections: Example Solution





- Pass a section through members FH, GH, and GI and take the right-hand section as a free body.

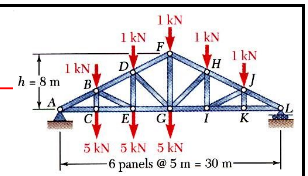
$$\tan\alpha = \frac{FG}{GL} = \frac{8\text{ m}}{15\text{ m}} = 0.5333 \qquad \alpha = 28.07^\circ$$

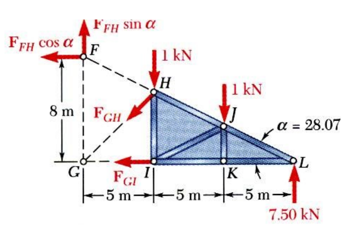


- Apply the conditions for static equilibrium to determine the desired member forces.

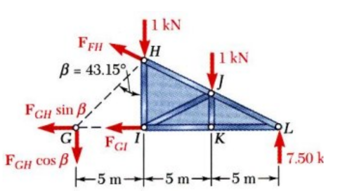
$$\sum M_H = 0$$
$$(7.50\text{ kN})(10\text{ m}) - (1\text{ kN})(5\text{ m}) - F_{GI}(5.33\text{ m}) = 0$$
$$F_{GI} = +13.13\text{ kN}$$
$$\boxed{F_{GI} = 13.13\text{ kN } T}$$

Method of Sections: Example Solution





$$\sum M_G = 0$$
$$(7.5\text{ kN})(15\text{ m}) - (1\text{ kN})(10\text{ m}) - (1\text{ kN})(5\text{ m})$$
$$+(F_{FH}\cos\alpha)(8\text{ m}) = 0$$
$$F_{FH} = -13.82\text{ kN}$$
$$\boxed{F_{FH} = 13.82\text{ kN } C}$$



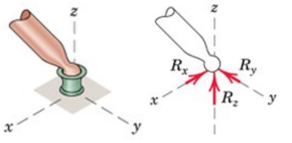
$$\tan\beta = \frac{GI}{HI} = \frac{5\text{ m}}{\frac{2}{3}(8\text{ m})} = 0.9375 \qquad \beta = 43.15^\circ$$
$$\sum M_L = 0$$
$$(1\text{ kN})(10\text{ m}) + (1\text{ kN})(5\text{ m}) + (F_{GH}\cos\beta)(15\text{ m}) = 0$$
$$F_{GH} = -1.371\text{ kN}$$
$$\boxed{F_{GH} = 1.371\text{ kN } C}$$

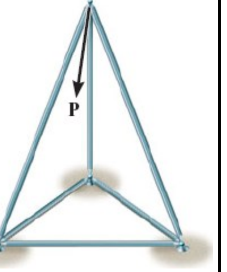
Structural Analysis: Space Truss

Space Truss

3-D counterpart of the Plane Truss

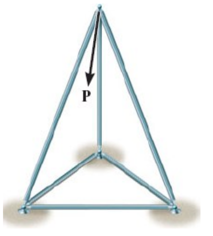
Idealized Space Truss → Rigid links connected at their ends by ball and socket joints



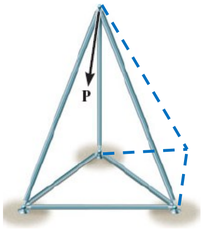


Structural Analysis: Space Truss

Space Truss



- 6 bars joined at their ends to form the edges of a tetrahedron as the basic non-collapsible unit
- 3 additional concurrent bars whose ends are attached to three joints on the existing structure are required to add a new rigid unit to extend the structure.



A space truss formed in this way is called a Simple Space Truss

If center lines of joined members intersect at a point
→Two force members assumption is justified
→ Each member under Compression or Tension

Structural Analysis: Space Truss

Static Determinacy of Space Truss

$\Sigma F_x = 0 \quad \Sigma M_x = 0$
$\Sigma F_y = 0 \quad \Sigma M_y = 0$
$\Sigma F_z = 0 \quad \Sigma M_z = 0$

Six equilibrium equations available to find out support reactions
→ if these are sufficient to determine all support reactions
→ The space truss is Statically Determinate Externally

Equilibrium of each joint can be specified by three scalar force equations
→ 3j equations for a truss with "j" number of joints
→ Known Quantities

For a truss with "m" number of two force members, and maximum 6 unknown support reactions → Total Unknowns = m + 6
("m" member forces and 6 reactions for externally determinate truss)
Therefore:
m + 6 = 3j→Statically Determinate Internally
m + 6 > 3j→Statically Indeterminate Internally
m + 6 < 3j→Unstable Truss

A necessary condition for Stability but not a sufficient condition since one or more members can be arranged in such a way as not to contribute to stable configuration of the entire truss

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Structural Analysis: Space Truss

Method of Joints

$\Sigma F_x = 0$
$\Sigma F_y = 0$
$\Sigma F_z = 0$

- If forces in all members are required
- Solve the 3 scalar equilibrium equations at each joint
- Solution of simultaneous equations may be avoided if a joint having at least one known force & at most three unknown forces is analysed first
- Use Cartesian vector analysis if the 3-D geometry is complex (∑**F**=0)

Method of Sections

$\Sigma F_x = 0 \quad \Sigma M_x = 0$
$\Sigma F_y = 0 \quad \Sigma M_y = 0$
$\Sigma F_z = 0 \quad \Sigma M_z = 0$

- If forces in only few members are required
- Solve the 6 scalar equilibrium equations in each cut part of truss
- Section should not pass through more than 6 members whose forces are unknown
- Cartesian vector analysis (∑**F**=0, ∑**M**=0)
- Not widely used

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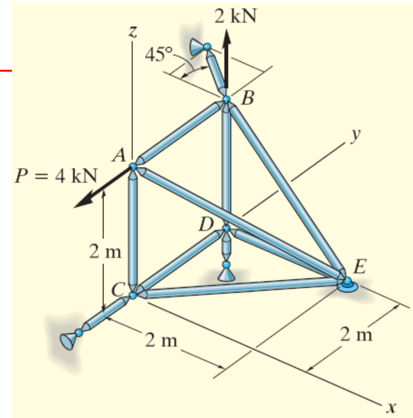
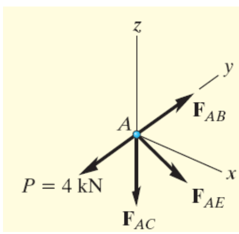
Space Truss: Example



Determine the forces acting in members of the space truss.

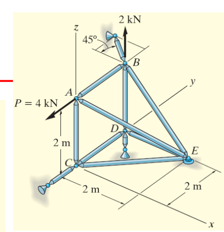
Solution:
Start at joint A: Draw free body diagram
Express each force in vector notation



$$\mathbf{P} = \{-4\mathbf{j}\}\text{ kN},\qquad \mathbf{F}_{AB} = F_{AB}\mathbf{j},\quad \mathbf{F}_{AC} = -F_{AC}\mathbf{k},$$

$$\mathbf{F}_{AE} = F_{AE}\left(\frac{\mathbf{r}_{AE}}{r_{AE}}\right) = F_{AE}(0.577\mathbf{i} + 0.577\mathbf{j} - 0.577\mathbf{k})$$

Space Truss: Example



For equilibrium,

$$\Sigma\mathbf{F} = \mathbf{0};\qquad \mathbf{P} + \mathbf{F}_{AB} + \mathbf{F}_{AC} + \mathbf{F}_{AE} = \mathbf{0}$$

$$-4\mathbf{j} + F_{AB}\mathbf{j} - F_{AC}\mathbf{k} + 0.577F_{AE}\mathbf{i} + 0.577F_{AE}\mathbf{j} - 0.577F_{AE}\mathbf{k} = \mathbf{0}$$

Rearranging the terms and equating the coefficients of **i**, **j**, and **k** unit vector to zero will give:

$$\Sigma F_x = 0;\qquad 0.577F_{AE} = 0$$

$$\Sigma F_y = 0;\qquad -4 + F_{AB} + 0.577F_{AE} = 0$$

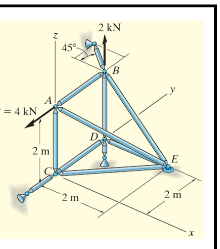
$$\Sigma F_z = 0;\qquad -F_{AC} - 0.577F_{AE} = 0$$

$$F_{AC} = F_{AE} = 0$$

$$F_{AB} = 4\text{ kN}$$

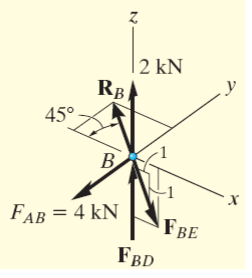
Next Joint B may be analysed.

Space Truss: Example



Joint B: Draw the Free Body Diagram

Scalar equations of equilibrium may be used at joint B



$$\Sigma F_x = 0; \qquad -R_B \cos 45^\circ + 0.707 F_{BE} = 0$$
$$\Sigma F_y = 0; \qquad -4 + R_B \sin 45^\circ = 0$$
$$\Sigma F_z = 0; \qquad 2 + F_{BD} - 0.707 F_{BE} = 0$$
$$R_B = F_{BE} = 5.66 \text{ kN} \quad \text{(T)}, \qquad F_{BD} = 2 \text{ kN}$$

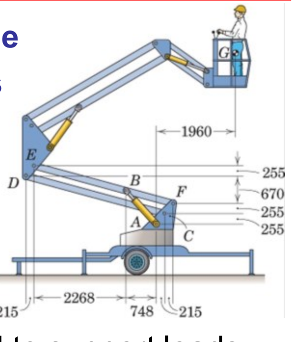
Using Scalar equations of equilibrium at joints D and C will give:

$$F_{DE} = F_{DC} = F_{CE} = 0$$

Frames and Machines

A structure is called a Frame or Machine if at least one of its individual members is a <u>multi-force member</u>

- member with 3 or more forces acting, or
- member with 2 or more forces and 1 or more couple acting



Frames: generally stationary and are used to support loads

Machines: contain moving parts and are designed to transmit and alter the effect of forces acting

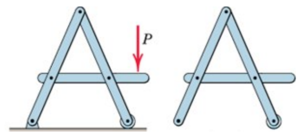
Multi-force members: the forces in these members in general will not be along the directions of the members

→ methods used in simple truss analysis cannot be used

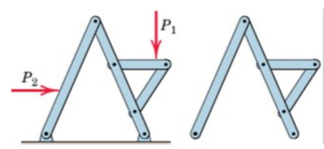
Frames and Machines

Interconnected Rigid Bodies with Multi-force Members

- Rigid Non-collapsible
  - –structure constitutes a rigid unit by itself when removed from its supports



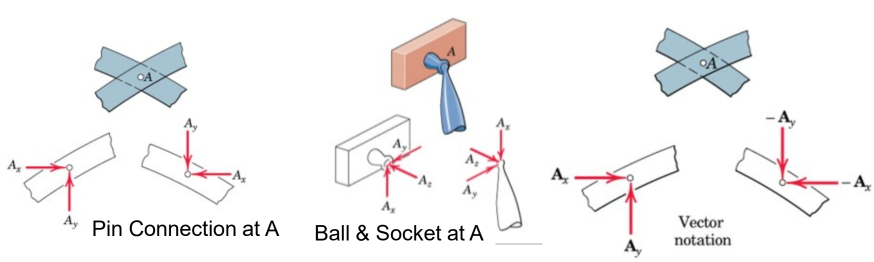
  - –first find all forces external to the structure treated as a single rigid body
  - –then dismember the structure & consider equilibrium of each part
- Non-rigid Collapsible
  - –structure is not a rigid unit by itself but depends on its external supports for rigidity
  - –calculation of external support reactions cannot be completed until the structure is dismembered and individual parts are analysed.



Frames and Machines

Free Body Diagrams: Forces of Interactions

- force components must be consistently represented in opposite directions on the separate FBDs (Ex: Pin at A).
- apply action-and-reaction principle (Ex: Ball & Socket at A).
- Vector notation: use plus sign for an action and a minus sign for the corresponding reaction



Pin Connection at A

Ball & Socket at A

Frames and Machines

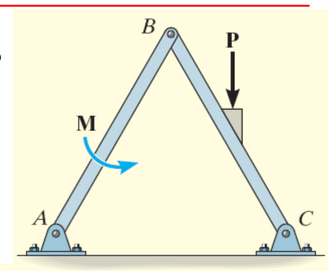
Example: Free Body Diagrams

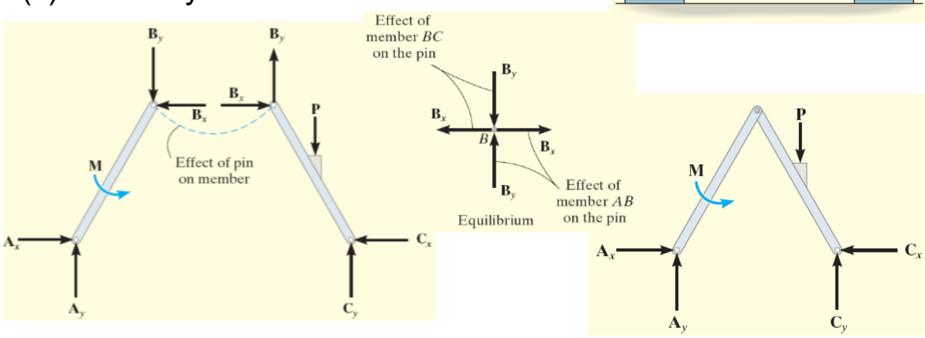
Draw FBD of

(a) Each member

(b) Pin at B, and

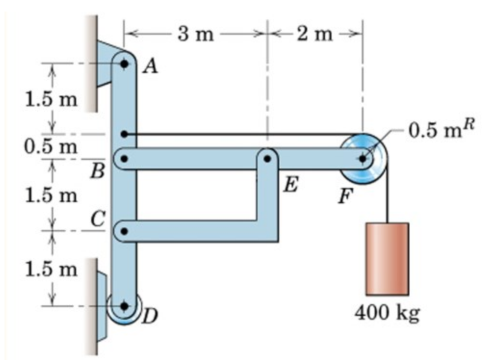
(c) Whole system





Frames and Machines

Example: Compute the horizontal and vertical components of all forces acting on each of the members (neglect self weight)



Frames and Machines

Example Solution:

3 supporting members form a rigid non-collapsible assembly

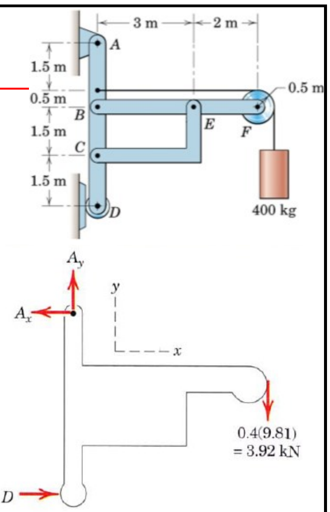
Frame Statically Determinate Externally

Draw FBD of the entire frame

3 Equilibrium equations are available

Pay attention to sense of Reactions

Reactions can be found out



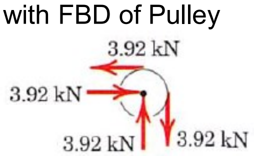
$[\Sigma M_A = 0]$	$5.5(0.4)(9.81) - 5D = 0$	$D = 4.32$ kN
$[\Sigma F_x = 0]$	$A_x - 4.32 = 0$	$A_x = 4.32$ kN
$[\Sigma F_y = 0]$	$A_y - 3.92 = 0$	$A_y = 3.92$ kN

Frames and Machines

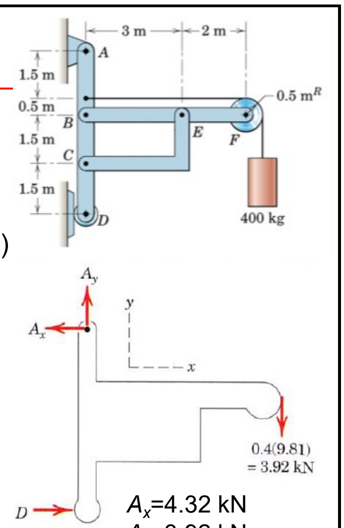
Example Solution: Dismember the frame and draw separate FBDs of each member

- show loads and reactions on each member due to connecting members (interaction forces)

Begin with FBD of Pulley



Then draw FBD of Members BF, CE, and AD

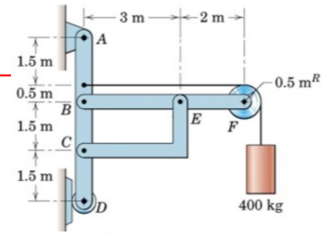


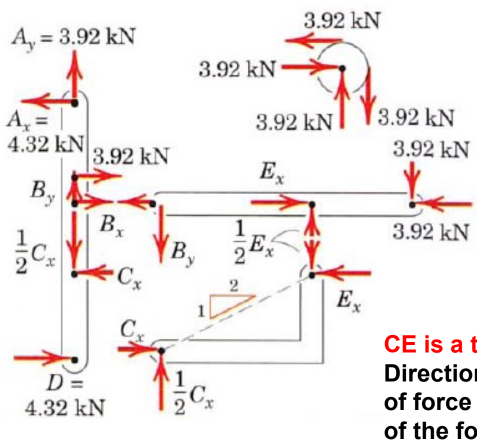
A_x=4.32 kN
A_y=3.92 kN
D=4.32 kN

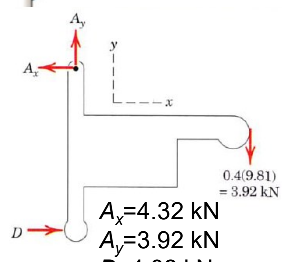
Frames and Machines

Example Solution:

FBDs







A_x=4.32 kN

A_y=3.92 kN

D=4.32 kN

CE is a two-force member

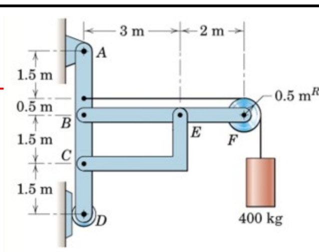
Direction of the line joining the two points of force application determines the direction of the forces acting on a two-force member.

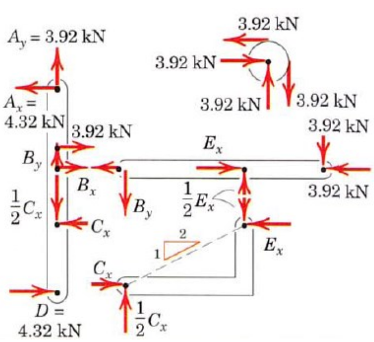
Shape of the member is not important.

Frames and Machines

Example Solution:

Find unknown forces from equilibrium





Member BF

$[\Sigma M_B = 0]$ $3.92(5) - \frac{1}{2}E_x(3) = 0$ $E_x = 13.08$ kN

$[\Sigma F_y = 0]$ $B_y + 3.92 - 13.08/2 = 0$ $B_y = 2.62$ kN

$[\Sigma F_x = 0]$ $B_x + 3.92 - 13.08 = 0$ $B_x = 9.15$ kN

Member CE

[∑Fx = 0] $C_x = E_x = 13.08$ kN

Checks:

$[\Sigma M_C = 0]$ $4.32(3.5) + 4.32(1.5) - 3.92(2) - 9.15(1.5) = 0$

$[\Sigma F_x = 0]$ $4.32 - 13.08 + 9.15 + 3.92 + 4.32 = 0$

$[\Sigma F_y = 0]$ $-13.08/2 + 2.62 + 3.92 = 0$

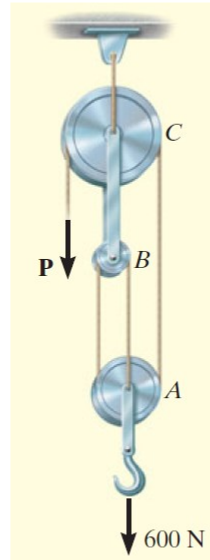
Frames and Machines

Example:

Find the tension in the cables and the force P required to support the 600 N force using the frictionless pulley system (neglect self weight)

Solution:

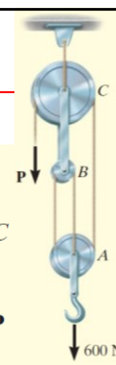
Draw the FBD



Frames and Machines

Example Solution:

Draw FBD and apply equilibrium equations



Pulley A

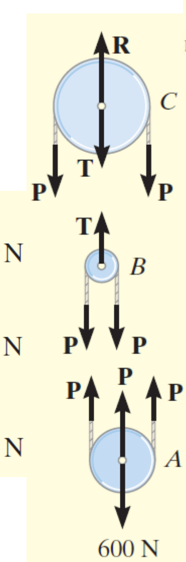
$+\uparrow \Sigma F_y = 0; \qquad 3P - 600\text{ N} = 0 \qquad P = 200\text{ N}$

Pulley B

$+\uparrow \Sigma F_y = 0; \qquad T - 2P = 0 \qquad T = 400\text{ N}$

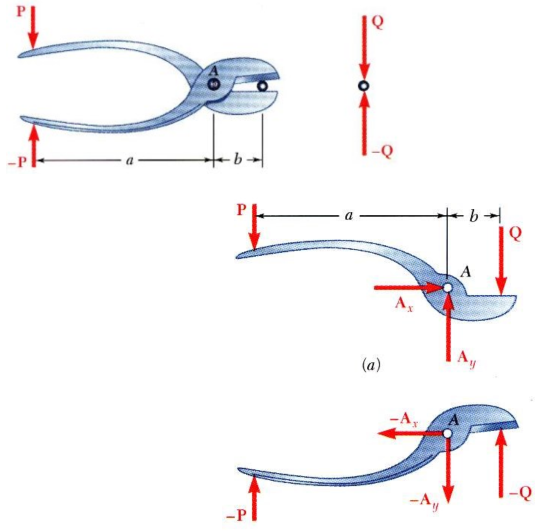
Pulley C

$+\uparrow \Sigma F_y = 0; \qquad R - 2P - T = 0 \qquad R = 800\text{ N}$

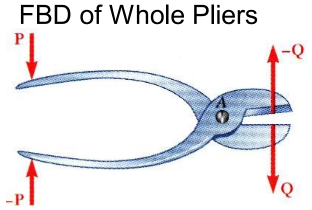


Frames and Machines

Example: Pliers: Given the magnitude of P, determine the magnitude of Q



FBD of Whole Pliers



FBD of individual parts

Taking moment about pin A

$Q=Pa/b$

Also pin reaction A_x=0

$A_y=P(1+a/b)$ OR $A_y=P+Q$

Frames and Machines

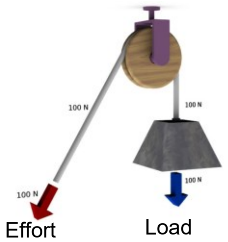
Definitions

- **Effort**: Force required to overcome the resistance to get the work done by the machine.
- **Mechanical Advantage**: Ratio of load lifted (*W*) to effort applied (*P*).
  Mechanical Advantage = *W/P*
- **Velocity Ratio**: Ratio of the distance moved by the effort (*D*) to the distance moved by the load (*d*) in the same interval of time.
  Velocity Ratio = *D/d*
- **Input**: Work done by the effort → Input = *PD*
- **Output**: Useful work got out of the machine, i.e. the work done by the load → Output = *Wd*
- **Efficiency**: Ratio of output to the input.

Efficiency of an ideal machine is 1. In that case, *Wd* =*PD* → *W/P*= *D/d.*

For an ideal machine, mechanical advantage is equal to velocity ratio.

Frames and Machines: Pulley System



Fixed Pulley

Effort = Load
→ Mechanical Advantage = 1

Distance moved by effort is equal to the distance moved by the load.
→ Velocity Ratio = 1

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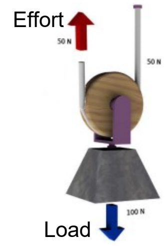
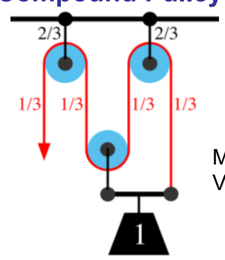


Movable Pulley

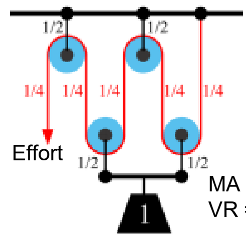
Effort = Load/2
→Mechanical Advantage = 2

Distance moved by effort is twice the distance moved by the load (both rope should also accommodate the same displacement by which the load is moved).
→ Velocity Ratio = 2

Compound Pulley

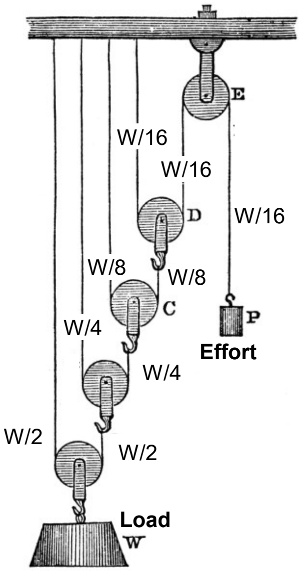


MA = 3
VR = 3



MA = 4
VR = 4

Frames and Machines: Pulley System



Compound Pulley

Effort required is 1/16th of the load.

Mechanical Advantage = 16.
(neglecting frictional forces).

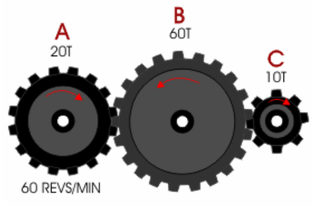
Velocity ratio is 16, which means in order to raise a load to 1 unit height; effort has to be moved by a distance of 16 units.

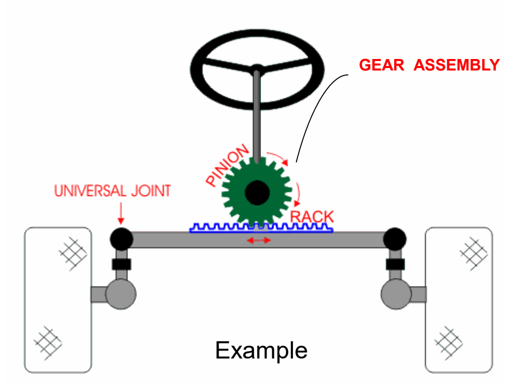
http://etc.usf.edu/

Frames and Machines

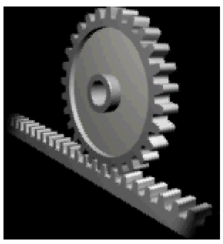
Gears

(Speed of Gear A x Number of teeth in Gear A) is equal to (Speed of Gear B x Number of teeth in Gear B) → Gear Ratio





Rack & Pinion

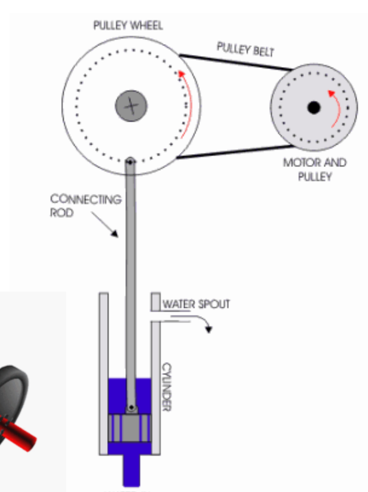


www.Petervaldivia.com

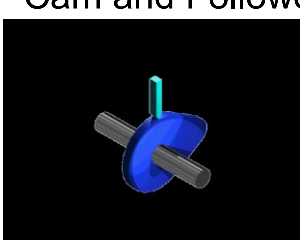
Frames and Machines

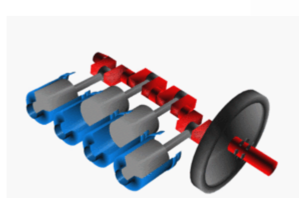
Gear and Belt

Crank Connecting Rod



Cam and Follower System

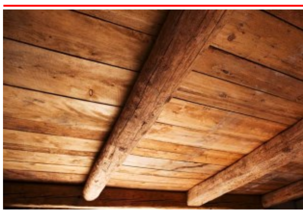


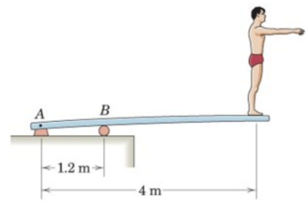


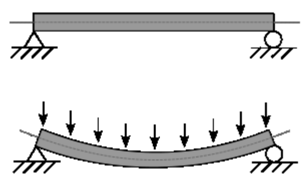
Example

www.Petervaldivia.com

Beams

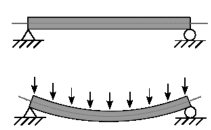






Beams are structural members that offer resistance to bending due to applied load

Beams



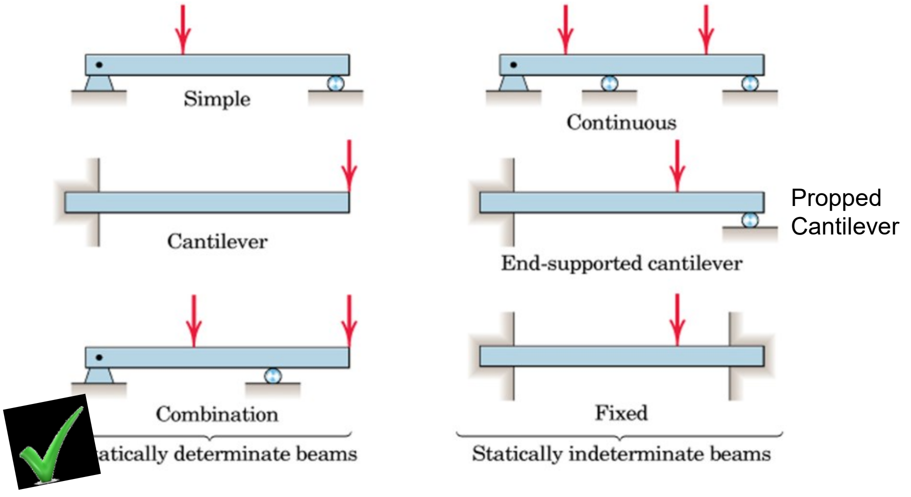
Beams

- mostly long prismatic bars
  - Prismatic: many sided, same section throughout
- non-prismatic beams are also useful
- cross-section of beams much smaller than beam length
- loads usually applied normal to the axis of the beam
- Determination of Load Carrying Capacity of Beams
- Statically Determinate Beams
  - Beams supported such that their external support reactions can be calculated by the methods of statics
- Statically Indeterminate Beams
  - Beams having more supports than needed to provide equilibrium

L08

Types of Beams

- Based on support conditions



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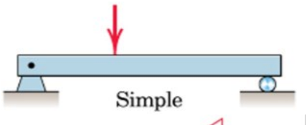
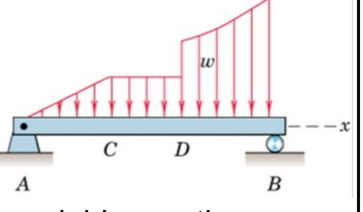
Types of Beams

- Based on type of external loading

Beams supporting Concentrated Loads



Beams supporting Distributed Loads



- Intensity of distributed load = w
- w is expressed as force per unit length of beam (N/m)
- intensity of loading may be constant or variable, continuous or discontinuous
  - discontinuity in intensity at D (abrupt change)
  - At C, intensity is not discontinuous, but rate of change of intensity (dw/dx) is discontinuous

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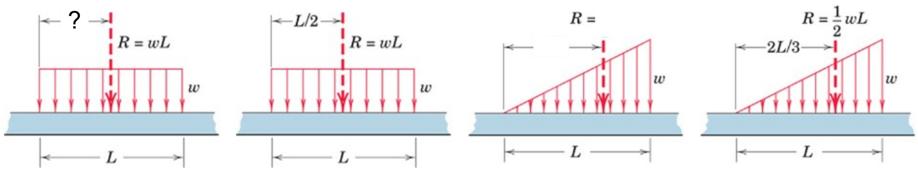
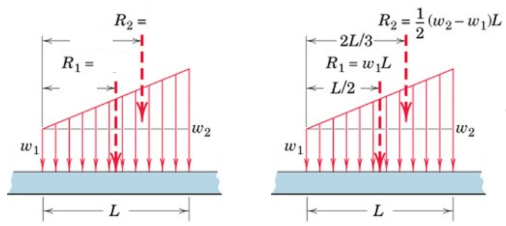
Beams

Distributed Loads on beams

- Determination of Resultant Force (*R*) on beam is important





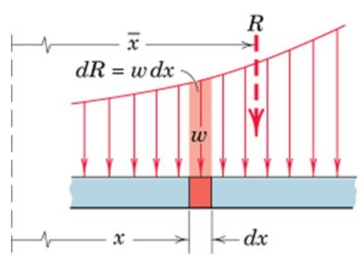
R = area formed by *w* and length *L* over which the load is distributed

R passes through centroid of this area

Beams

Distributed Loads on beams

- General Load Distribution



Differential increment of force is
$dR = w\,dx$

Total load *R* is sum of all the differential forces

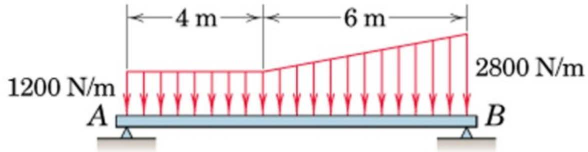
$R = \int w\,dx$ acting at centroid of the area under consideration

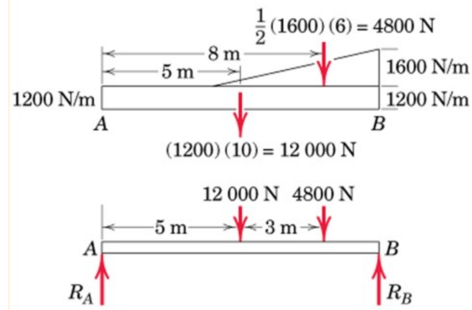
$$\bar{x} = \frac{\int xw\,dx}{R}$$

Once *R* is known reactions can be found out from Statics

Beams: Example

Determine the external reactions for the beam





R_A = 6.96 kN, R_B = 9.84 kN

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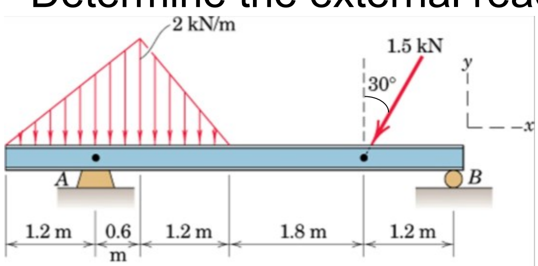
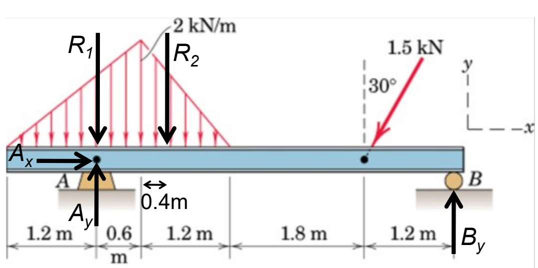
Beams: Example

Determine the external reactions for the beam



Dividing the un-symmetric triangular load into two parts with resultants R_1 and R_2 acting at point A and 1m away from point A, respectively.

R_1 = 0.5x1.8x2 = 1.8 kN
R_2 = 0.5x1.2x2 = 1.2 kN



$\sum F_x=0 \rightarrow$ $\boldsymbol{A_x}$ = 1.5sin30 = **0.75 kN**

$\sum M_A=0 \rightarrow$
4.8xB_y = 1.5cos30x3.6 + 1.2x1.0
$\boldsymbol{B_y}$ **= 1.224 kN**

$\sum F_y=0 \rightarrow$
A_y = 1.8+1.2+1.5cos30-1.224
$\boldsymbol{A_y}$ **= 3.075 kN**

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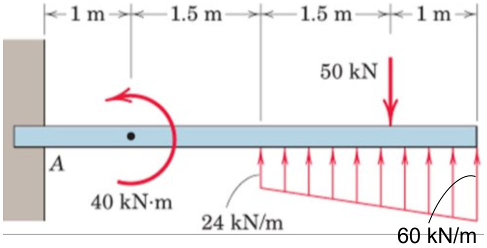
Beams: Example

Determine the external reactions for the beam



Dividing the trapezoidal load into two parts with resultants R_1 and R_2

R_1 = 24x2.5 = 60 kN @ 3.75m → A
R_2 = 0.5x2.5x36=45 kN@4.17m → A
(distances from A)

$\sum M_A=0$ →
M_A-40+50x4.0-60x3.75-45x4.17=0
M_A = 253 kNm

$\sum F_y=0$ →
A_y -50+60+45 = 0
A_y = -55 kN → Downwards